

Ayush Kumar

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EDUCATION

National Research Tomsk State University
Tomsk, Russia

2023 - Present

*Specialist Degree in Mathematics and Computer Science (Equivalent to M.Sc.),
Faculty of Mathematics and Computer Science (02.03.01)
Specializing in **Computational Mathematics and Computer Modeling***

Core coursework and professional focus:

- machine learning: Applied ML and artificial neural networks (Python, C++), ML fundamentals, and deep learning technologies.
- computational mathematics: advanced numerical methods, computer modeling, and numerical solutions for industrial mathematics.
- high-performance computing: parallel programming technologies and parallel computing methods.
- mathematical modeling: differential equations and finite-difference schemes applied to ecology, medicine, and mechanics.
- applied science: computer-based analysis of natural science problems and computational mechanics.
- computer workshop on mechanics;
- models and numerical methods in natural science problems.

EXPERIENCE

Wunder Fund RNN

Machine Learning Researcher (Independent)

Sep 2025- Jan 2026

- Designed and deployed a production-grade inference pipeline for High-Frequency Trading (HFT) market state forecasting, achieving a Top 30 Global Ranking in the Wunder Fund Challenge
- Solved extreme deployment constraints (Single-core CPU, <20MB storage limit) by engineering a custom INT8 Dynamic Quantization pipeline in PyTorch. This reduced model memory footprint by 75%, allowing for the deployment of a 10-model ensemble within the strict resource budget.
- Architected a "Hive Mind" Ensemble consisting of Deep Residual GRUs and LSTMs with Mish activations and Squeeze-and-Excitation blocks. This architecture maximized predictive performance while minimizing inference latency.
- Implemented robust MLOps practices, including automated artifact serialization, custom loss function implementation (Huber, Pinball), and rigorous cross-validation strategies to prevent overfitting on small-sample financial datasets.

Turing AI

Mathematical Problem Creator & Programmer

- Created original, hard-level mathematical and algorithmic problems (similar to Project Euler) to help train AI models in reasoning and logic.
- Wrote efficient reference solutions in **C++** and **Python**, ensuring they run within strict time and memory limits ($O(n \log n)$ complexity).
- Designed rigorous test cases and checking for edge cases to verify that the solutions are correct and stable.
- Wrote detailed, step-by-step explanations and hints to clearly show the mathematical logic behind each solution.

PROJECTS

[AuthCore – Custom Authentication & RBAC System](#) [GitHub](#)

- AuthCore - Custom Authentication and Role-Based Access Control (RBAC) System
- Developed a secure and stateless authentication system in **Python**, leveraging **JSON Web Tokens (JWT)** for efficient session management.
- Implemented robust password security using **bcrypt** for hashing.
- Designed and built a fine-grained **Role-Based Access Control (RBAC)** model to safeguard **RESTful API** endpoints.
- Created custom middleware to validate user permissions against a **PostgreSQL** database.
- Automated deployment and maintenance workflows through Shell scripting
- Delivered a **scalable, efficient, and secure backend solution** tailored to meet modern application security needs.

[Efficient Neural HFT \(Market State Forecasting\)](#) [GitHub](#) | [Hugging Face](#) (demo)

- **Architected a "Hive Mind" Ensemble** consisting of Deep Residual GRUs and LSTMs, optimized with Mish activations and Squeeze-and-Excitation blocks to maximize predictive power in financial time-series forecasting.
- **Engineered a custom INT8 Dynamic Quantization pipeline** in PyTorch to meet extreme deployment constraints of <20MB storage and single-core CPU execution.

- **Reduced model memory footprint by 75%**, allowing a sophisticated 10-model ensemble to operate within a strictly limited resource budget.
- **Implemented automated MLOps workflows**, including artifact serialization and custom financial loss functions (Huber, Pinball) to improve model robustness.
- **Developed a production-grade inference pipeline** specifically designed for high-frequency trading (HFT) latency and performance standards.

CERTIFICATIONS

- [Deep Learning AI Specialization](#) – *DeepLearning.AI (January 2024)*
- [Neural Networks and Deep Learning](#) – *DeepLearning.AI (December 2023)*
- [IBM Machine Learning](#) – *Coursera (July 2023)*
- [Python Pro Bootcamp \(100 Days of Code\)](#) – *Udemy (December 2022)*
- [Full-Stack Web Development Bootcamp](#) – *Udemy (May 2022)*

SKILLS

- **Machine Learning:** Deep Learning, LLMs, CNNs, NLP, Data Preprocessing
- **Frameworks & Tools:** PyTorch, TensorFlow, LangChain, Scikit-learn, Pandas
- **Programming:** Python, SQL, Java, C++
- **Concepts:** Model Evaluation & Validation, Feature Engineering, Hyperparameter Tuning, Prompt Engineering, Transfer Learning & Fine-Tuning, Algorithm Design & Analysis, MLOps Principles, System Design

ADDITIONAL

- **Languages:** English (Fluent), Russian (Intermediate)

Awards and Honors

- **Top 28 Global Ranking** in the Wunder Fund RNN Challenge 2025—Machine Learning Competition for market state forecasting.
- **3rd Place Winner** in a coding Hackathon Code Gym by Т-Банк Образование (April 2025).
- AvitoTech ML CUP 2025 by Avito hosted by Open Data Science(ods.ai) **Awarded Diploma for Successful Challenge Completion.**